

Yingtian Shi

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Education

Georgia Institute of Technology

Aug 2024 – Present

PhD in Computer Science

- **Research Area:** Human-centered AI, Human AI Co-evolution, LLMs, Computer Vision, Sensing
- Advised by Prof. Thomas Ploetz.

Tsinghua University

Aug 2020 – June 2023

MS in Computer Science & Technology

- **Research Area:** Multimodal Interaction, Human Behavior Understanding, Sensing
- Advised by Prof. Yuanchun Shi.

Tsinghua University

July 2016 – June 2020

BS in Computer Science & Technology

Technical Skills

Programming Languages: Python, C++, C, C#, Java, SQL, CSS/HTML, Typescript

Machine Learning: PyTorch, TensorFlow, scikit-learn, OpenCV

Experience

Graduate Research Assistant [08/2025 – Current]

Atlanta, GA

AI Caring, Georgia Institute of Technology

- Pioneered an AI framework by integrating multi-modal sensor fusion from nearable and wearable devices to build high-fidelity activity understanding. By leveraging Graph Neural Networks (GNNs) and temporal pattern mining, the system identifies complex activity sequences to equip agents with deep contextual awareness, enabling proactive decision-making and personalized assistance within smart home environments.

Research Assistant [05/2025 - 08/2025]

Beijing, China

Tsinghua University

- Engineered a scalable Multi-Agent System (MAS) platform designed for intelligent agent recommendation and sophisticated collaborative interactions. I implemented an autonomous task understanding layer and an output integration module to synthesize multi-agent results, resulting in a system capable of handling collaboration of different agents; this work is currently under peer review for a top-tier conference.

Graduate Research Assistant [08/2024 - 04/2025]

Atlanta, GA

Georgia Institute of Technology

- Developed a high-precision Computer Vision pipeline utilizing Visual Foundation Models (e.g., SAM2) and YOLO for the automated detection of diabetic foot ulcers. To ensure clinical AI fairness, I applied Data Augmentation and Few-shot Learning techniques to maintain diagnostic accuracy across diverse skin tones and lighting conditions, culminating in a published paper at IEEE BHI 2025.

Research Assistant [10/2023 - 06/2024]

Beijing, China

Tsinghua University

- Built a domain-specific Vision-Language Model (VLM) and an expert alignment platform to automate hazard identification on construction sites. The system employs Instruction Tuning (LoRA) and automation integration to bridge the gap between raw AI visual recognition and specialized human safety protocols, allowing agents to rapidly interpret and apply complex safety regulations.

Visited Student Researcher [06/2023 – 09/2023]

Stanford, CA

Stanford University

- Developed a comprehensive developer ecosystem comprising a Python-based SDK and a custom VSCode

Extension designed to automate multimodal interaction matching and application feature discovery. I engineered a specialized Domain-Specific Language (DSL) with an Object-Oriented framework to formalize interaction logic, enabling LLM Agents to autonomously parse, map, and execute complex UI tasks. By integrating these agents directly into the IDE, the platform facilitates real-time semantic discovery and seamless interaction matching, significantly reducing the cognitive load for developers building multimodal applications.

Selected Publications

[CHI 25] GenieWizard: Multimodal App Feature Discovery with Large Language Models [\[Paper\]](#) [↗](#)

Jackie (Junrui) Yang, **Yingtian Shi**, Chris Gu, Zhang Zheng, Anisha Jain, Tianshi Li, Monica S. Lam, James A. Landay

[CHI 25] Enhancing Smartphone Eye Tracking with Cursor-Based Interactive Implicit Calibration [\[Paper\]](#) [↗](#)

Chang Liu, Xiangyang Wang, Chun Yu, **Yingtian Shi**, Chongyang Wang, Ziqi Liu, Chen Liang, Yuanchun Shi

[CHI 24] ReactGenie: A Development Framework for Complex Multimodal Interactions Using Large Language Models [\[Paper\]](#) [↗](#)

Jackie (Junrui) Yang, **Yingtian Shi**, Yuhan Zhang, Karina Li, Daniel Wan Rosli, Anisha Jain, Shuning Zhang, Tianshi Li, Monica S. Lam, James A. Landay

[IMWUT 23] Understanding In-Situ Programming for Smart Home Automation [\[Paper\]](#) [↗](#)

Xiaoyi Liu*, **Yingtian Shi***, Chun Yu, Cheng Gao, Tianao Yang, Chen Liang, Yuanchun Shi

[IEEE VR 23] ConeSpeech: Exploring Directional Speech Interaction for Multi-Person Remote Communication in VR (**Best Paper Honorable Mention Award**) [\[Paper\]](#) [↗](#)

Yukang Yan, Haohua Liu, **Yingtian Shi**, Jingying Wang, Ruici Guo, Zisu Li, Chun Yu, Xuhai Xu, Yuntao Wang, Yuanchun Shi

[CHI 21] Facesight: Enabling hand-to-face gesture interaction on ar glasses with a downward-facing camera vision [\[Paper\]](#) [↗](#)

Yueting Weng, Chun Yu, **Yingtian Shi**, Yuhang Zhao, Yukang Yan, Yuanchun Shi

[IMWUT 20] Headcross: Exploring head-based crossing selection on head-mounted displays [\[Paper\]](#) [↗](#)

Yukang Yan*, **Yingtian Shi***, Chun Yu, Yuanchun Shi

Mentored Students

Masters: Abivishaq Balasubramanian, Ankith N Chandra, Yeonsu Shin, Kunyu Gao

Undergraduates: Xuyang Lu, Yuhan Wang, Jinda Yang, Haoyu Li